Learn C++ Programming -Beginner to Advance- Deep Dive in C++

Section 5:C++ Basics

Lecture 25. Primitive Data Types

1. Introduction

Data types are available in C++ language.

Data types are needed because every language supports data types. Without data types, there is no use of writing programs.

Data types are used by declaring variables.

---

2. Categories of Data Types

Data types in C++ are categorized into three main types:

Data Types
- **Primitive**
- **User Defined**
- **Derived**

---

3. Primitive Data Types

Primitive data types are the basic data types of C++.

Primitive means they are directly provided by the compiler.

Primitive data types are categorized into:

Primitive
- **Integral**
- **Boolean**
- **Floating Point**

---

4. Integral Type

Integral means there is no decimal point.

Under integral type:

Integral
├── int
└── char

int and char come under integral type because they do not have a decimal point.

---

5. Boolean Type

Boolean means:

true
false

---

6. Floating Point Type

Floating point means data with decimal point.

Decimal is also called floating point.

Under floating point:

Floating Point
├── float
└── double

float and double are similar, but some differences are there.

---

7. User Defined Data Types

User defined data types are:

enum

structure
union
class

enum means enumeration.

C++ supports object orientation.

Actual C++ concepts and programming start from classes.

---

8. Derived Data Types

Derived data types are:

Array
Pointer
Reference

Arrays, pointers, and references are separate detailed topics.

---

9. Data Types and Their Sizes

Data types discussed are:

int
float
double
char
bool

Sizes mean how many bytes of memory they take.

Data Type	Size
int	2 bytes or 4 bytes
float	4 bytes
double	8 bytes
char	1 byte
bool	Undefined, can be 1 byte or less

---

10. Range of Values

Each data type has a possible range of values depending on its size.

Important ranges to remember:

Data Type	Range
int	-32768 to 32767
char	-128 to 127
float	-3.4 × 10^-38 to 3.4 × 10^38
double	-1.7 × 10^-308 to 1.7 × 10^308

The range of int and char should be remembered.

For float and double, remembering is good, but not a problem if not remembered exactly.

---

11. Integer Size

int may take either 2 bytes or 4 bytes depending on the compiler.

Most compilers now take int as 4 bytes.

Turbo C takes int as 2 bytes because it runs in DOS.

For explanation, int is considered as 2 bytes.

---

12. How int Gets Its Range

If int takes 2 bytes:

2 bytes = 16 bits

One bit is reserved for sign.

This bit is called the most significant bit.

If the sign bit is:

1 → negative number
0 → positive number

Remaining bits are used for storing the number.

2^15 = 32768

So the range becomes:

-32768 to 32767

The positive side is:

0 to 32767

The negative side is also considered, but negative zero is not a number.

So negative zero is taken as:

-32768

That is why there is one extra number on the negative side.

---

13. Character Size and Range

char takes:

1 byte

1 byte means:

8 bits

One bit is used as sign bit.

Remaining 7 bits are used for storing the number.

2^7 = 128

So the range of char is:

-128 to 127

---

14. Characters and ASCII Codes

Computer works on binary number system, that is:

0s and 1s

Characters are also represented using 0s and 1s.

For characters, codes are given.

These codes are called:

ASCII codes

ASCII means:

American Standard Code for Information Interchange

For every alphabet, letter, and special symbol, there is a code.

---

15. Important ASCII Codes

Character	ASCII Code
A	65
B	66
Z	90
a	97
b	98
z	122
0	48
1	49
9	57

These codes should be remembered.

---

16. Modifiers

Modifiers are used to modify data types as per requirement.

Two modifiers discussed are:

unsigned
long

---

17. unsigned Modifier

Example:

unsigned int

Normal integer supports both negative and positive numbers.

unsigned int takes only positive numbers.

For 2-byte integer, the range becomes:

0 to 65535

The sign bit is not there.

All 16 bits are used for storing the number.

If negative numbers are not needed and a bigger range is required, unsigned int can be used.

---

18. unsigned char

Example:

unsigned char

The range becomes:

0 to 255

unsigned can be used only with:

int

char

It cannot be used with every data type.

---

19. long Modifier

long can be used with integer.

Example:

long int

If int takes 2 bytes, then long int takes 4 bytes.

If int takes 4 bytes, then long int takes 8 bytes.

This depends on the compiler.

---

20. long double

long can also be used with double.

Example:

long double

long double takes:

10 bytes

long float has no meaning because long float itself is double.

---

21. Modifiers Cannot Be Used with Everything

unsigned cannot be used with everything.

long cannot be used with everything.

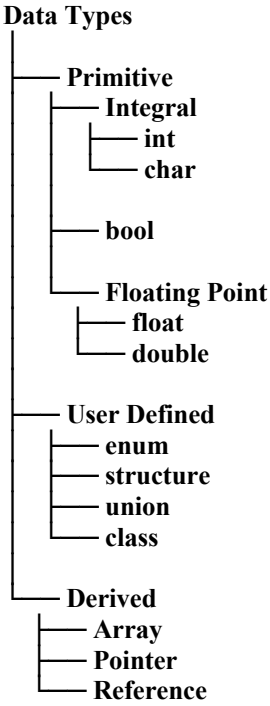
Example:

long char

is not used.

---

22. Complete Data Type Chart



---

23. Key Points

Data types are available in C++.

Primitive data types are basic data types of C++.

Primitive data types are directly provided by the compiler.

Primitive data types are integral, boolean, and floating point.

Integral means no decimal point.

Floating point means decimal point.

int and char come under integral type.

float and double come under floating point type.

User defined data types are enum, structure, union, and class.

Derived data types are array, pointer, and reference.

Each data type takes some memory.

Each data type has a range of values.

Characters are represented using ASCII codes.

Modifiers discussed are unsigned and long.